

Hankel-Structured Channel Estimation for Rydberg Atomic MIMO Receivers

Results digest

Abdullah Haatim · Department of Electrical and Electronics Engineering
BITS Pilani — Pilani Campus

Short version. All numbers are traceable; the full derivations, methodology and provenance are in the master document. Three experiment families appear here and are *never* pooled — each number is tagged [B3] [EXT] or [TD] (§6).

1. The problem, in three sentences

A Rydberg atomic receiver reads a radio field through an optical transition whose response tracks field *magnitude*. So the receiver measures

$$\mathbf{Z} = |\mathbf{GS} + \mathbf{B} + \mathbf{W}|,$$

where $\mathbf{G}$ is the $N \times K$ channel we want, $\mathbf{S}$ the pilots, $\mathbf{B}$ a known reference field and $\mathbf{W}$ noise — **phase is lost, and the noise sits inside the modulus**. Multi-user channel estimation is therefore a phase retrieval problem, and the standard baseline is Gerchberg–Saxton with an EM refinement (EM-GS).

2. The idea, in one picture

For a uniform linear array each channel column is a sum of L complex exponentials in the element index. Stack it into a Hankel matrix and that sum becomes a rank- L matrix:

$$\mathcal{H}(\mathbf{g}) = \sum_{\ell=1}^L \alpha_{\ell} \mathbf{v}_{\ell} \mathbf{w}_{\ell}^{\top} \implies \text{rank } \mathcal{H}(\mathbf{g}) \leq L.$$

So **HS-GS** = EM-GS with a low-rank Hankel projection inserted after every update. Two things make it non-trivial:

- **The prior has a capacity.** An N -element array can express at most $r_{\max}(N) = \lceil N/2 \rceil$ ranks. If $L \geq r_{\max}$ the constraint says nothing. That is why the method’s value depends on aperture.
- **The order $\hat{L}$ must be chosen without an oracle.** It is selected from held-out pilots. When the selector returns $\hat{L} \geq r_{\max}$ the projection switches off — the method *abstains* rather than impose structure it cannot support.

With the projection disabled, HS-GS reproduces EM-GS to $\max|\text{diff}| = 0$. The projection is provably the only difference between them.

3. Headline results

Table 1 is the whole story in one place. If you read nothing else, read the first four rows: the method works, and *how well* it works is governed by array size, in a way the next section explains.

4. The three findings that matter

1. The gain is controlled by how much of the prior’s capacity the channel uses — not by array size or path count on their own. A controlled sweep confirms the mechanism directly: hold everything fixed and vary only L , and the gain decays monotonically from +7.04 dB

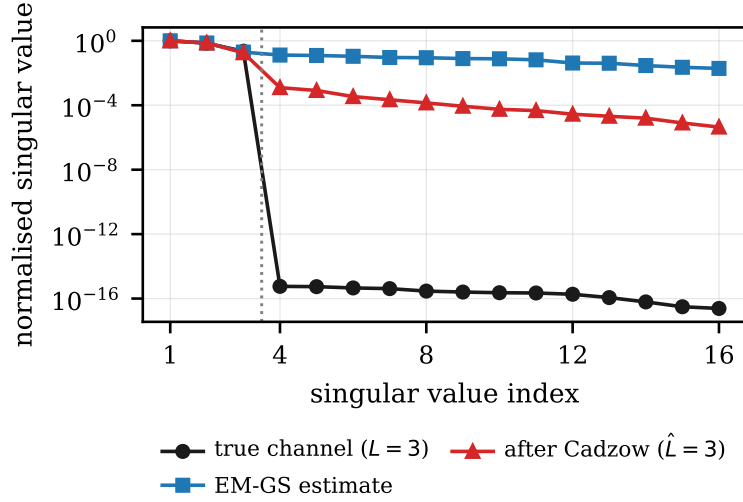


Figure 1: Why the prior exists, and what the projection actually achieves. [EXT] Hankel singular values for one channel column ($N = 32$, $L = 3$). The true channel is exactly rank L ($\sigma_4/\sigma_1 = 5.7 \times 10^{-16}$). The EM-GS estimate is full numerical rank (0.130) — noise fills the spectrum. Four Cadzow sweeps restore the cliff, but only to 1.2×10^{-3} : the structure is imposed *approximately*, which caps how much of the available gain the method can capture. One illustrative realisation.

Result	Value	Family
Gain of HS-GS over EM-GS at $N = 32$, $P = 30$	+2.27 to +3.04 dB	B3
Aperture scaling, $N = 8/16/32$	-0.19 / +0.78 / +2.85 dB	B3
Fraction of trials HS-GS wins, $N = 8/16/32$	33.5 / 74.0 / 95.3%	B3
Constraint active, $N = 8/16/32$	57.5 / 92.7 / 99.6%	B3
EM-GS vs. Cramér–Rao bound, $\text{SNR} \geq 5$ dB	within 0.051 dB	B3
Constrained bound below unconstrained one	7.05–7.11 dB	B3
Path-count decay, $L = 2 \rightarrow 16$	+7.04 $\rightarrow$ -0.12 dB	EXT
EM-GS is flat in both N and L	0.012 / 0.191 dB	B3 / EXT
Effective-rank zero crossing, $N = 16/32/64$	0.588 / 0.518 / 0.544	TD
Prospective prediction error, gen. 1 / gen. 2	0.13 / 0.315 dB	TD
Gain invariance in user count K , $\text{SNR} \geq 5$ dB	0.095 dB spread	TD

Table 1: The numbers worth remembering. The two flatness figures are the control: EM-GS ignores structure, so it should not move when structure changes — and it does not.

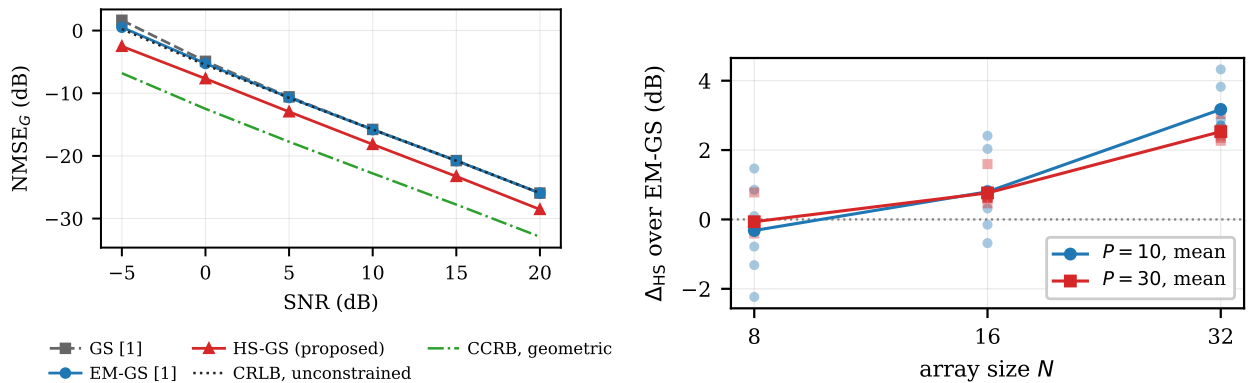


Figure 2: Left: the main result. [B3] NMSE vs. SNR at $N = 32$, $K = 3$, $P = 30$, 400 paired trials per point. HS-GS beats EM-GS by 2.27–3.04 dB across the sweep. EM-GS itself tracks the unconstrained CRLB to within 0.051 dB above 5 dB, which is what makes the benchmark meaningful rather than vacuous. **Right: the gain depends on aperture.** [B3] It is *negative* at $N = 8$, where capacity $r_{\max} = 4$ is smaller than the typical path count.

at $L = 2$ to -0.12 dB at $L = 16 = r_{\max}$ [EXT]. EM-GS does not move (0.191 dB across the whole sweep), which is exactly what an estimator that ignores structure should do.

2. Raw path count is the wrong coordinate; effective rank is the right one. Replacing L by the effective rank of the Hankel matrix ($r_{\text{eff}} = \exp(-\sum p_i \ln p_i)$, normalised by capacity) collapses three different array sizes onto one curve, crossing zero at 0.588/0.518/0.544 for $N = 16/32/64$ [TD]. The *location* of the boundary is array-invariant to within 0.07.

3. The coordinate predicts, not just describes. Two channel generators never used to build the relation were run, with predictions committed to version control *before* measurement and with no refitting. Errors: 0.13 dB and 0.315 dB [TD].

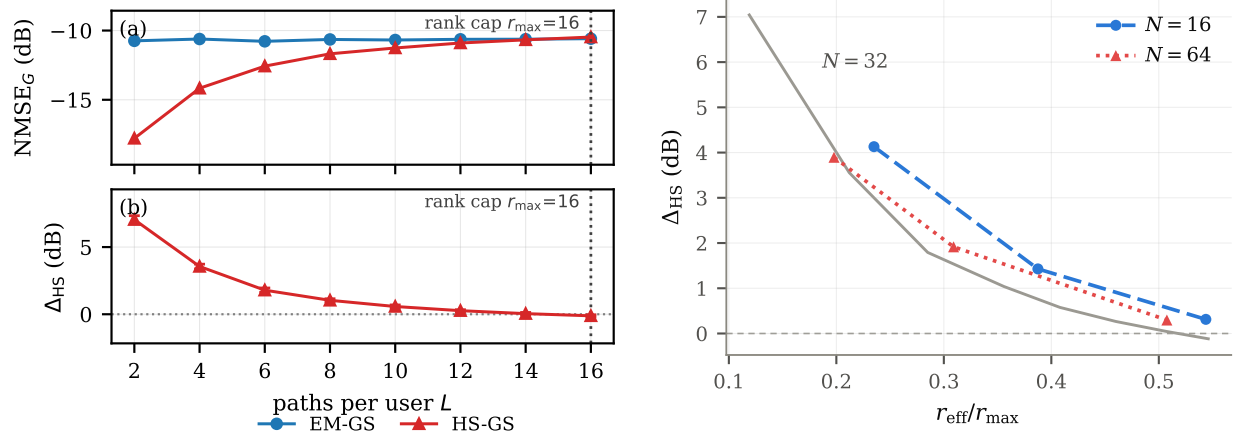


Figure 3: Left: the controlled experiment. [EXT] Vary only the path count L . The gain decays to zero exactly as L approaches capacity $r_{\text{max}} = 16$; EM-GS stays flat. The hypothesis was recorded before the run. **Right: the collapse.** [TD] Plotted against effective rank over capacity, three array sizes land on one curve and cross zero at nearly the same place.

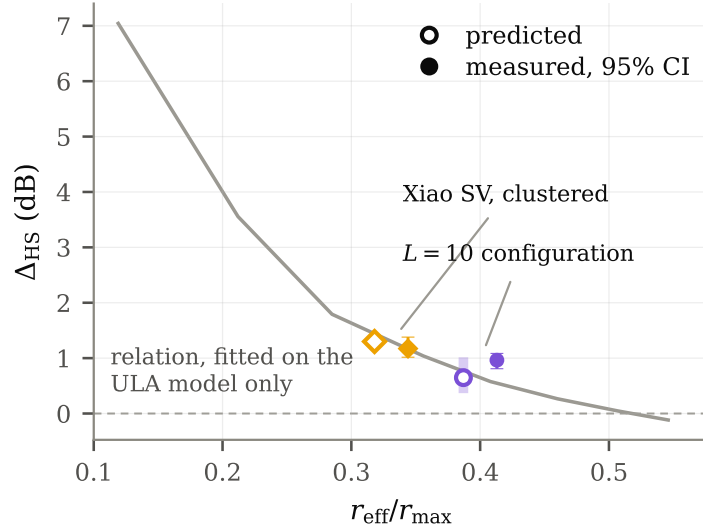


Figure 4: Predictions registered before measurement. [TD] Hollow markers are what the relation predicted; filled markers with bars are what was measured, with 95% confidence intervals. The relation was fitted on the ULA model only and was *not* refitted for either generator. The prediction rule is a piecewise-linear interpolation on a fixed 8-point table — no regression, no spline, so it can be re-executed exactly.

5. What does not work

Stated plainly, because these bound how far the result travels.

- $N = 8$ is a mixed result, not a small positive one. The gain is +1.47 dB at -5 dB SNR but falls to -2.23 dB at high SNR, and the projection is inactive in 42.5% of trials [B3]. Truncation

bias does not shrink with noise, so as SNR rises the trade stops paying and then reverses. The sign of any single scalar summary at $N = 8$ depends on the pooling.

- **The collapse is imperfect.** The *location* of the boundary is array-invariant; the gain *magnitude* is not. A one-signed residual of mean 0.493 dB (max 0.901) between $N = 16$ and $N = 64$ is unexplained [TD].
- **A pre-registered criterion narrowly failed.** The K -invariance test misses its own falsifier by 0.006 dB when pooled over all SNR (it passes comfortably at $\text{SNR} \geq 5$ dB) [TD]. Reported, not dropped.
- **The second prediction is an order of magnitude worse than the first** (0.315 vs. 0.13 dB), though still inside its stated interval.
- **The bounds do not formally govern the estimators plotted against them.** CRLB and CCRB constrain *unbiased* estimators; every estimator here is biased. They are informative benchmarks, not laws these curves must obey.
- **Simulation only**, geometric ULA channels only, and order selection costs 57.7–85.5% of runtime with no attempt yet to reduce it.

6. The three experiment families

These are **different experiments**. Numbers from one are not comparable to numbers from another as measurements of the same thing.

Tag	Store	Configuration
B3	results/track.b/b3/	$N \in \{8, 16, 32\}$, $P \in \{10, 30\}$, 6 SNR points, 400 trials/point, 4 Cadzow sweeps. The principal experiment.
EXT	trackB_hankel_emgs/	$P = 30$ only, 7 SNR points, 600/400/200 trials. A separate diagnostic run.
TD	results/track.d/partB9/	$N \in \{16, 32, 64\}$, $P = 20$, adaptive order, 1 Cadzow sweep. One sweep and four sweeps define different operators.

B3 and EXT both measure aperture gain and *disagree* ($-0.19/+0.78/+2.85$ vs. $+0.03/+0.81/+2.45$ dB) because they are different experiments. Both are correct for their own configuration; neither should be quoted for the other.

Full document with derivations, algorithms, methodology and complete limitations:
 paper/master/haatim_hsgs_master.pdf (13 pp.). Verification of every number against its store:
 paper/master/MASTER.PAPER.AUDIT.md. No simulation was rerun to produce this digest.